

Artificial Intelligence (AI)

Lab Sheet No: 2

Introduction to First Order Predicate Logic (FOPL)

The use of symbolic logic to represent knowledge is not new in that it predates the modern computer by a number of decades. Even so, the application of logic as a practical means of representing and manipulating knowledge was not demonstrated until the early 1960s. Today First Order Predicate Logic (FOPL) or predicate calculus has assumed one of the important roles in AI for representing the knowledge.

The understanding of FOPL for AI student has several benefits. One, logic offers the formal approach to reasoning that has a sound theoretical foundation. Next, the structure of FOPL is flexible enough to permit the accurate representation of the natural language reasonably well.

1. The predicates of FOPL have precisely the same syntactic structure as they have in Prolog - i.e. a predicate name with terms as arguments, each term being a

-constant,

-variable, or

-function with term(s) as argument(s).

2. Variables must be quantified by either
 $\forall$ (Universal Quantifier) meaning "for all" or
 $\exists$ (Existential Quantifier) meaning "there exists"

e.g $\exists x P(x)$ means "some unspecified constant has property P". Each quantifier has a SCOPE which is defined as the textual area in which it binds occurrences of its variable. This scope is usually delimited by brackets, unless it is obvious as in the example above.

These extensions critically extend the expressive power of logic. Consider the following sentences which are given a natural corresponding syntax in predicate calculus:

Natural Language	Predicate
John is the brother of Jim.	brother_of(john,jim).
Socrates is dead and buried.	dead(socrates)&buried(socrates).

All men are mortal.	$\forall x(Man(x) \square Mortal(x))$
Ram loves all animals.	$\forall xAnimals(x) \square Loves(ram, x)$
Poppy is a dog.	$Dog(Poppy)$
Grandparent is a parent of one's parent	$\forall x, yGrandparent(x, y) \Leftrightarrow \exists zParent(x, z) \cap Parent(z, y)$
Parent and child are inverse relation.	$\forall x, yParent(x, y) \Leftrightarrow Child(y, x)$

Rules combine facts to increase knowledge of the system

son(X,Y):-

male(X),child(X).

X is a son of Y if X is male and X is a child of Y.

Program 1: Write the following statements in FOPL form and by converting them into prolog program test the given goal.

1. Every American who sells weapons to hostile nations is a criminal.
2. Every enemy of America is a hostile.
3. Iraque has some missiles.
4. All missiles of Iraque were sold by George.
5. George is an American.
6. Iraque is a country.
7. Iraque is the enemy of America.
8. Missiles are weapens.

PROGRAM

PREDICATES

hostile (STRING)

enemy_of_america (STRING)

american (STRING)

criminal (STRING)

```
sells_missiles (STRING, STRING)
has_missile (STRING)
country (STRING)

CLAUSES
criminal (X) :-
    american (X), sells_missiles (X, Y), hostile (Y) .
enemy_of_america (X) :-
    hostile (X) .
enemy_of_america ("Iraq") .

hostile (X) :-
    country (X) .
has_missile ("Iraq") .
sells_missiles ("George", "Iraq") .
american ("George") .
country ("Iraq") .

GOAL
criminal ("George") .
```

Program 2: Monkey- Banana Problem

Monkey-Banana Problem is the famous problem in AI. Where there is a room containing a monkey, a chair, and bananas that have been hung from the center of the ceiling of the room; out of reach from monkey. If the monkey is clever enough, he can reach the bananas by placing the chair directly below the bananas and climbing on the top of the chair.

Now the problem is to use FOPL to represent this monkey-banana problem and prove that monkey can reach the bananas.

The program is given below. Before running the program, think carefully what are the essential objects of the problem and how should them be arranged in predicate logic.

```
PREDICATES
in_room (symbol)
```

```
dexterous(symbol)
tall(symbol)
can_move(symbol,symbol,symbol)
can_reach(symbol,symbol)
get_on(symbol,symbol)
can_climb(symbol,symbol)
close(symbol,symbol)
under(symbol,symbol)
can_climb(symbol,symbol)

CLAUSES
in_room(bananas).
in_room(chair).
in_room(monkey).
dexterous(monkey).
tall(chair).
can_move(monkey,chair,bananas).
can_climb(monkey,chair).

can_reach(X,Y):-
    dexterous(X),close(X,Y).

close(X,Z):-
    get_on(X,Y),
    under(Y,Z),
    tall(Y).
get_on(X,Y):-
    can_climb(X,Y).

under(Y,Z):-
    in_room(X),
    in_room(Y),
    in_room(Z),
    can_move(X,Y,Z).

GOAL
can_reach(monkey,apple)
```

Program 2: Write the following statements in FOPL form and by converting them into prolog program test the given goal.

9. Every American who sells weapons to hostile nations is a criminal.
10. Every enemy of America is a hostile.
11. Iraque has some missiles.
12. All missiles of Iraque were sold by George.
13. George is an American.
14. Iraque is a country.
15. Iraque is the enemy of America.
16. Missiles are weapons.

PROGRAM

PREDICATES

```
hostile (STRING)
enemy_of_america (STRING)
american (STRING)
criminal (STRING)
sells_missiles (STRING, STRING)
has_missile (STRING)
country (STRING)
```

CLAUSES

```
criminal (X) :-
    american (X), sells_missiles (X, Y), hostile (Y) .

enemy_of_america (X) :-
    hostile (X) .
enemy_of_america ("Iraq") .

hostile (X) :-
    country (X) .
has_missile ("Iraq") .
```

```
sells_missiles("George", "Iraq").  
american("George").  
country("Iraq").
```

GOAL

```
criminal("George").
```

Assignments: Write the following statements in FOPL form and by converting them into prolog program test the different goals.

1. Assume the following facts:

- Steve only likes easy courses
- Science courses are hard
- All the courses in the basket weaving department are easy
- BK301 is a basket weaving course

Use resolution to answer the question, "What course would Steve like?"

2. Write the following statements in FOPL form and by converting them into prolog program test the different goals.

1. Horses are mammals.
2. An offspring of a horse is a horse.
3. Bluebeard is a Charlie's parent.
4. Offspring and parents are inverse relations.
5. Every mammal has a parent.

Now check, **is Charlie a horse?**

3. Write the following statements in FOPL form and by converting them into prolog program test the different goals.

Consider the following “**Student Evaluation System**”

Student Evaluation System

Rule I

If there are chances that a student will get good position
AND there is no other student with marks greater than him
Then he will get first position.

Rule II

If Student attends all his lectures regularly
AND Studies his course books
Then He will cover his course contents

Rule III **If** a Student cover his course contents.

Then there are chances that he will get good position.

Suppose we know the following facts,

- i. A student attends all his lectures regularly
- ii. A student studies his course books
- iii. There is no one with marks higher than him

Show that this student will get first position using forward and backward chaining.